InCaRisk user guide

1. Problem

How can the *InCaRisk* tool be used to estimate the probabilities and number of additional cancer cases resulting from inhalation exposure?

2. Solution: method and approach

In practice, health risk assessment (HRA) is often difficult to implement because multiple risk indicators are available (Table 1), which can lead to divergent results and, therefore, to different prevention recommendations and measures depending on the studies and countries [Petit et al. 2019]. Even when the same studies and the same critical health effects are considered, health agencies may issue different recommendations because of differences in methodology, policy, and expert judgment. This can be problematic because decision-making in this area has a direct and significant impact on the prevention and protection of the health of populations exposed to hazardous agents, as well as on the incidence of the associated diseases.
Regarding the HRA of populations exposed to carcinogenic chemical agents, there is currently no gold standard for selecting and using a risk indicator depending on the health context and exposure scenario encountered. At this level, a standardized approach would be needed to provide consistent results, enable direct comparisons across studies and countries, and assess health impacts if changes, developments, and/or preventive actions were to occur. The interactive web application *InCaRisk* (for Inhalation Cancer Risks) was developed to address these issues [Petit et al. 2020; Petit and Bicout, 2022].

Tableau 1: Existing cancer risk indicators/formulas

Indicator	Abbreviation	Formula	Reference
Modified lung cancer risk	LCR	$LCR = 1 - e^{-(C \times IUR)}$	[Callen 2014, Hsu 2014, Jia 2011, Lin 2008, Liu 2010, Ramirez 2011, Wang 2012]
Modified cancer risk	RC	$RC = 1 - e^{-\left(\frac{C \times ET \times EF \times ED \times IUR}{AT}\right)}$	[Irvine 2014]
Modified incremental lifetime risk	ILCR	$ILCR_1 = 1 - e^{-\left(\frac{C \times CSF \times \sqrt[3]{BW/70} \times EF \times ED \times TI}{BW \times AT \times PEF}\right)}$ $ILCR_2 = 1 - e^{-\left(\frac{C \times CSF \times \sqrt[3]{BW/70} \times EF \times ED \times TI \times cf}{BW \times AT}\right)}$ $ILCR_3 = 1 - e^{-\left(\frac{C \times CSF \times TI \times EF \times ED \times ET}{BW \times AT}\right)}$ $ILCR_4 = 1 - e^{-\left(\frac{C \times CSF \times TI \times EF \times ED \times A}{BW \times AT}\right)}$ $ILCR_5 = 1 - e^{-\left(\frac{C \times CSF \times TI \times EF \times ED \times cf}{BW \times AT}\right)}$	[Huang 2016, Soltani 2015] [Chen 2006, Hoseini 2016, Singh 2016, Wang 2014] [Hu 2007, Li 2013, Qu 2015] [Li 2014] [Yu 2015]
Modified risk	RM	$RM = 1 - e^{-\left(\frac{C \times (TI_a \times EF \times ED \times ET) \times (IUR \times BW \times cf)}{BW \times AT \times TI_b}\right)}$	[Amarillo 2014]

A: inhalation absorption factor, A: averaging time, BW: body mass, C: chemical concentration, cf: conversion factor, CSF: inhalation cancer slope factor, ED: exposure duration, EF: exposure frequency, IUR: inhalation unit risk, ET: exposure time, PEF: particles emission factor, TI: inhalation rate, TI_a: inhalation rate in m^3/hour, TI_b: inhalation rate in m^3/day.

InCaRisk is an interactive tool for estimating cancer risks after inhalation exposure to carcinogenic chemicals (Figure 1). Available free of charge on the website https://exporisk-timc.imag.fr/InCaRisk/, and accessible from any web browser, *InCaRisk* is regularly updated and offers the following features:

(i) interactive visualization of cancer risk probabilities (among 64 different cancer types) and the expected number of cancer cases in an exposed population, based on a given exposure scenario to one chemical substance among 509 available and one health agency among 18 available;

(ii) comparison of cancer risks obtained with 8 existing indicators and a consensus indicator based on the aggregation of the 8 available indicators (Table 1);

(iii) comparison of cancer risks obtained according to different health agencies and an inter-agency indicator;

(iv) a database linking chemical substances to cancer types.

All generated results (figures and tables), as well as a report summarizing the results obtained with *InCaRisk*, can be downloaded.

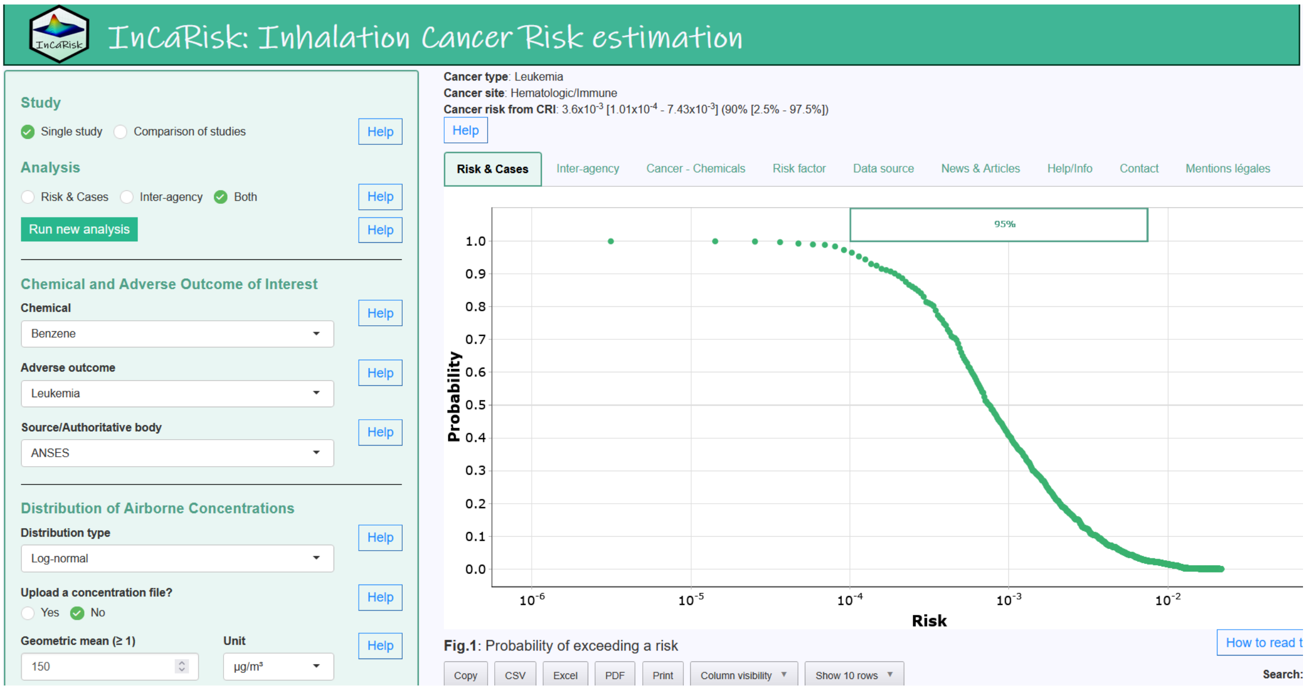

Figure 1: *InCaRisk* screenshot

InCaRisk requires an initialization time that depends on your internet connection, your computer's processing power, and the number of users online in real time.

InCaRisk consists of an area for defining the exposure scenario (green box on the left side of the screen) and an area for viewing the results and other relevant information through 9 tabs (white section on the right side of the screen).

2.1 Exposure scenario

Help buttons are placed next to each parameter to explain what it refers to and how to set it. Simply click the help buttons to display the explanations. The first parameter to define is the type of study to be conducted: a single study or a comparison of studies. In comparison of studies, multiple populations and/or exposure scenarios can be compared. The second choice concerns the type of analysis, with either the use of a unit risk estimate (URE) from a given health agency, comparison of the cancer risk estimates based on all available UREs, or both. The "*run new analysis*" button allows you to launch a new analysis. If this button has not been clicked, the analysis and the selected exposure scenario will not be launched or considered.

The following choices concern the chemical substance of interest, the toxic effect selected (i.e., cancer type), and the selection of a health agency's URE. You then either upload a file containing the concentrations of the pollutant of interest measured in the atmosphere, or define a concentration distribution, starting with the type of distribution (log-normal by default), the mean and unit of the concentrations, the standard deviation of the distribution, the number of simulated measured values, the possible inclusion of concentration fluctuations over time, the number of simulations to be performed, and the type of modeling approach to use (Monte Carlo or artificial intelligence-based). The greater the number of simulations, the more precise the estimate, but the longer the computation time. Note that the wider the standard deviation, the less reliable the estimate, which highlights the importance of having similar exposure groups (SEGs).

You then need to define the exposure scenario, namely the number of hours per day, days per year, and total number of years of exposure, as well as the population size (i.e., number of exposed individuals). The following choices concern the risk indicators (Table 1) and the health agencies to be compared.

2.2 File upload

It is possible to upload concentration files for single studies, whereas this is mandatory for study comparisons. To upload a file, simply click the "*Import data*" buttons. A pop-up window will then open, allowing you to select the file to upload (Figure 2). After selecting the file, click the gear icon to define the decimal separator if necessary. ***Note that only the period (.) is accepted***. Use the "*View*" tab to preview the dataset and verify that the information is correct. The "Update" tab allows you to modify the dataset. Do not forget to click "*Apply changes*" to save the changes. Finally, click "*Import data*" to load the selected file.

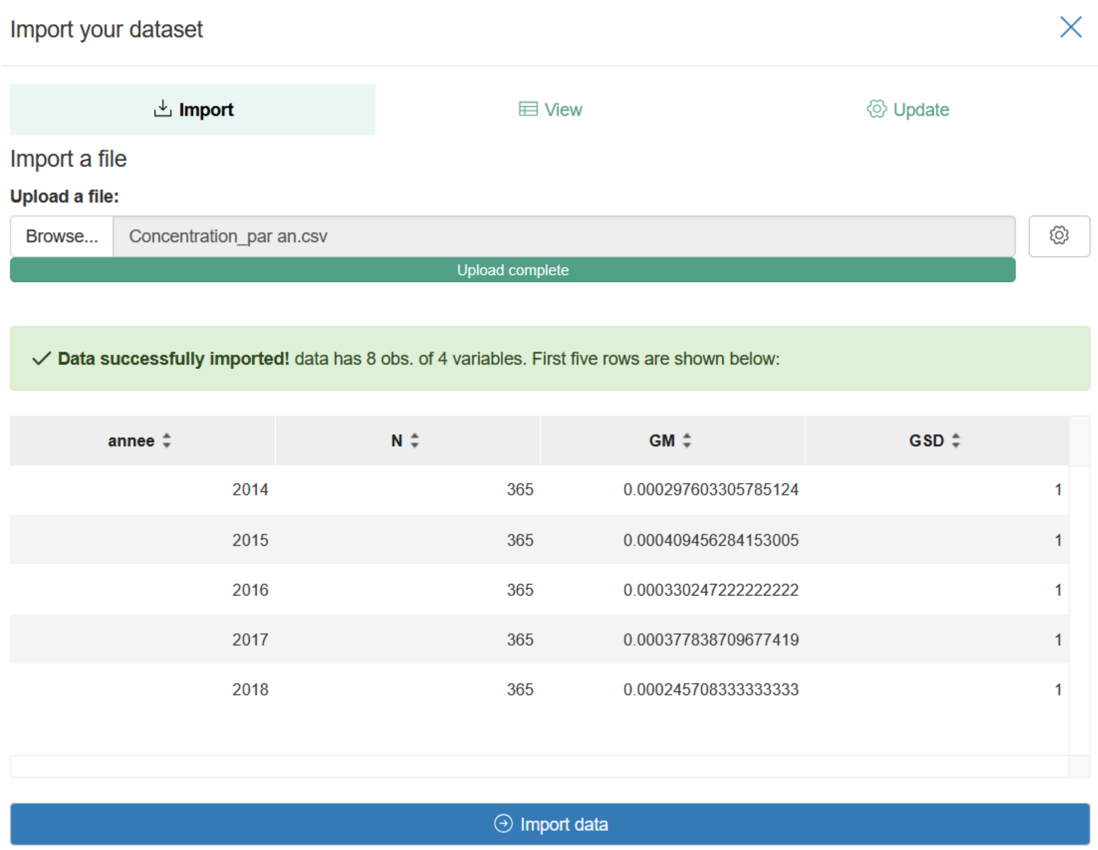

Figure 2: File upload in *InCaRisk*

2.3 Results and additional information

All results are generated using the consensus risk indicator [Petit et al. 2021]. Above the tabs, a reminder is displayed of the cancer type (e.g., leukemia), its location (e.g., lung), and the estimated additional probability of cancer occurrence for the exposure scenario.

The first tab (*Risk & Cases*) provides the probability and number of additional cancer cases resulting from inhalation for the defined exposure scenario. All tables and figures are interactive. By hovering over a figure, icons appear that allow interaction with the graph (e.g., zoom, export). As for the tables, they can also be manipulated and exported. Blue help buttons provide further explanations. The first figure shows the probability of exceeding a given risk threshold (e.g., 10-4) (Figure 3).

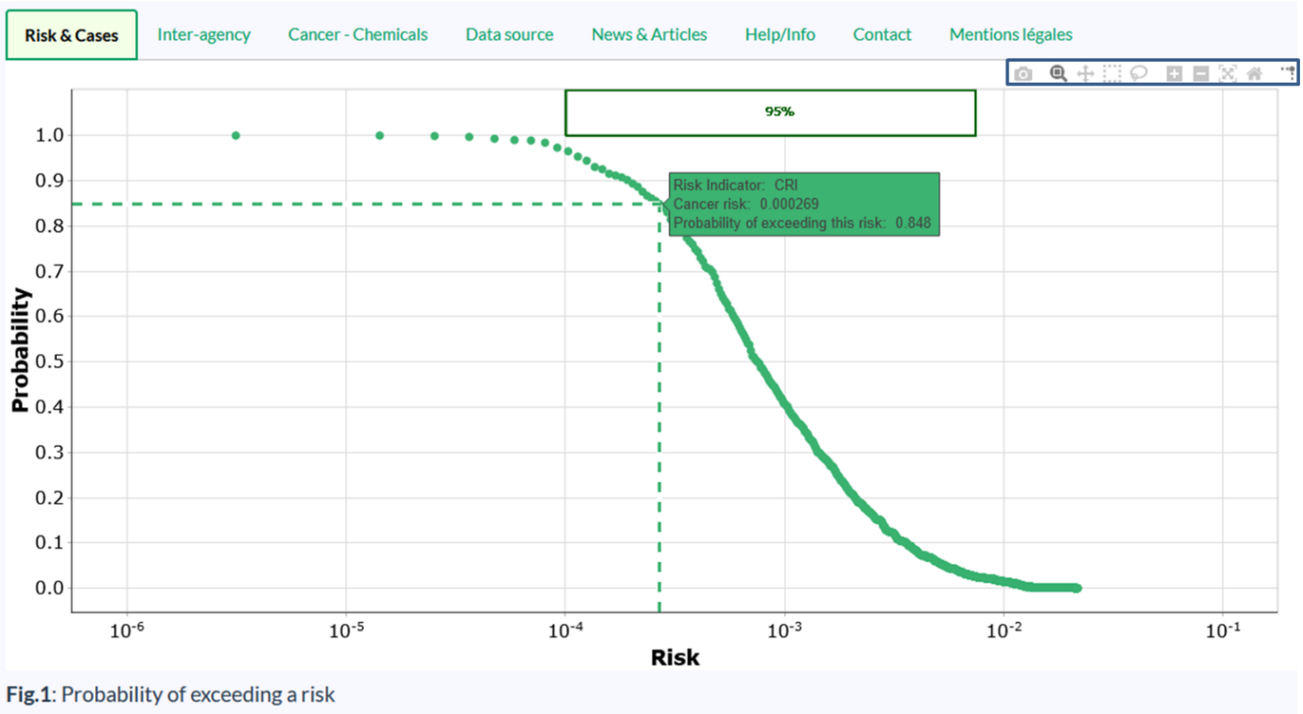

Figure 3: Probability of exceeding risk

The second figure shows the risk distribution as a boxplot and violin plot (a symmetrical/mirror representation of the shape of the distribution) (Figure 4).

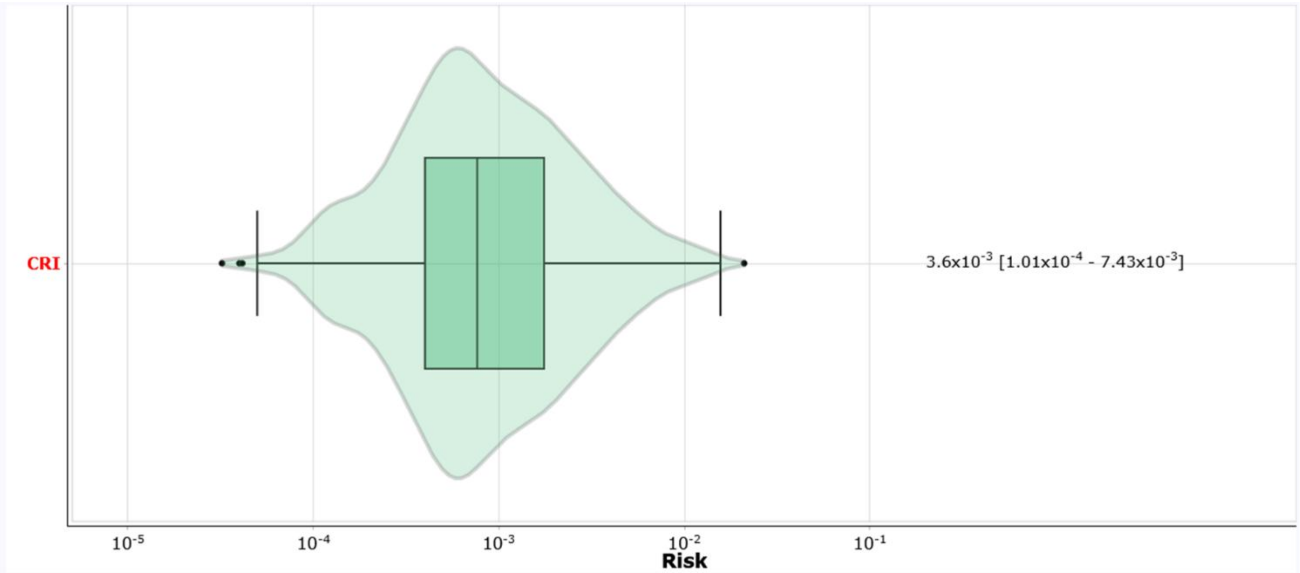

Fig.2: Distribution of the estimated cancer risk

Figure 4: Cancer risk distribution

The third figure provides an estimate of the uncertainty associated with the cancer risk estimate, while the last figure shows the distribution of the expected number of additional cancer cases in the exposed population (Figure 5).

Expected number of additional cancer cases per year from CRI: 3.78x10^{1} [1.06 - 7.81x10^{1}] (90% [2.5% - 97.5%])

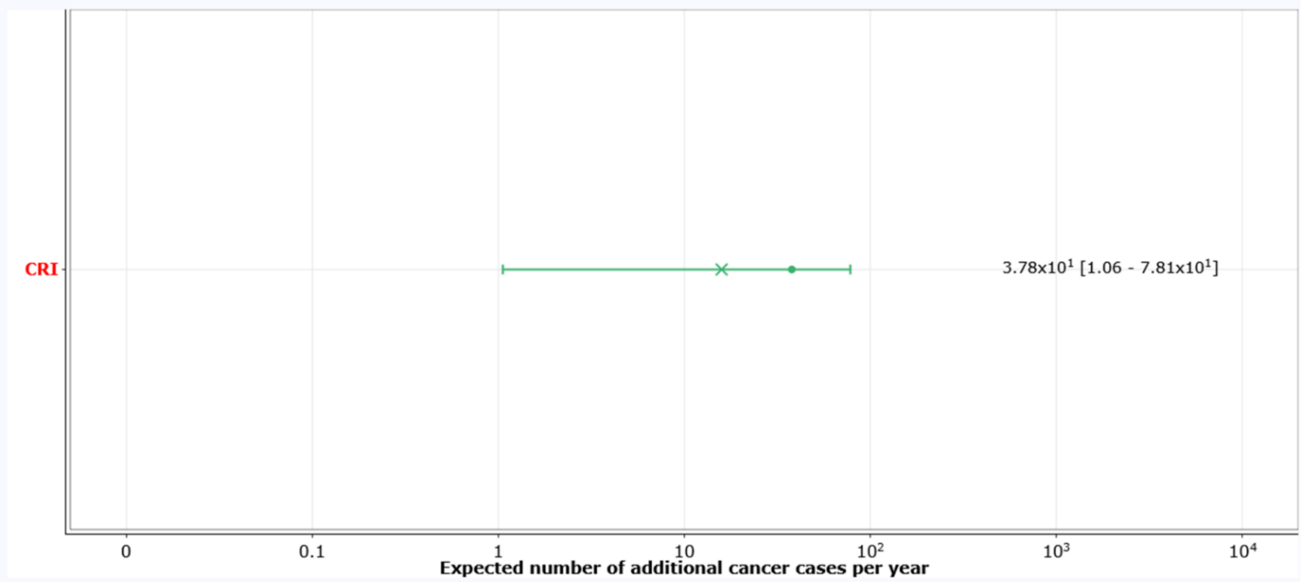

Fig.4: Estimation of the expected number of additional cancer cases per year

X: Mean
o: 90th percentile
|: 95th confidence interval

Figure 5: Distribution of additional cancer cases

Exercise

Consider an individual working in a coke plant and exposed to benzo[a]pyrene (BaP). Assume that this individual is exposed under the following scenario:

• *Concentration*: geometric mean BaP = 31 ng/m3 and geometric standard deviation = 4.
• *Exposure*: 7 hours per day, 270 days per year for 40 years.
• *Toxic effect*: lung cancer.
• *Health agency*: WHO URE.
• *Exposed population*: 10,000 individuals.

Questions

1. What is this individual's cancer risk?
Answer: excess lifetime risk (ELR) = 3.03x10-3 [3.78x10-5; 7.42x10-3]

2. Is this risk acceptable according to EU recommendations?
Answer: No because ELR > 10-4

3. What is the probability of exceeding the risk value of 1x10-3?
Answer: approximately 29%

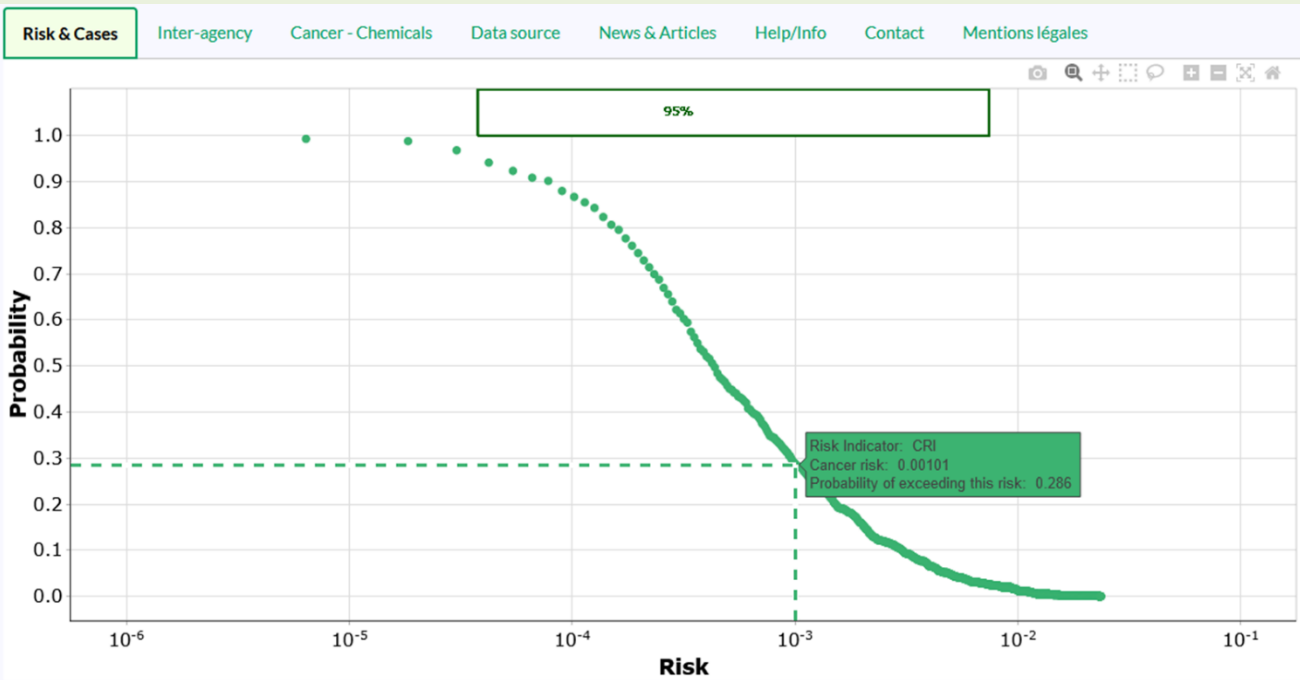

Fig.1: Probability of exceeding a risk

4. How many additional cancer cases are expected?
Answer: approximately 30 cases

The second tab (*Inter-agency*) makes it possible to compare risk estimates according to the UREs of different health agencies and thus measure their level of agreement (Figures 6 and 7). There can sometimes be a factor of 10 between two health agencies! The choice of health agency is therefore essential and has a major impact on cancer risk estimation and the interpretation of the results.

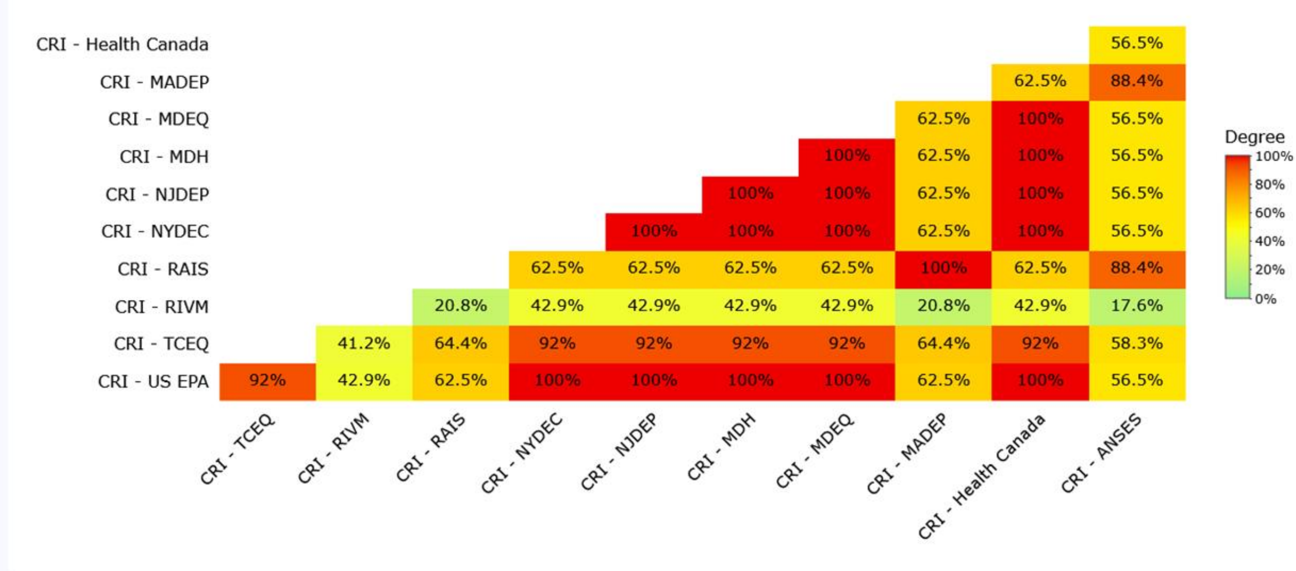

Fig.5: Matrix of degrees of agreement between authoritative bodies

Figure 6: Matrix of degrees of agreement between IUR-based risk distributions among health agencies

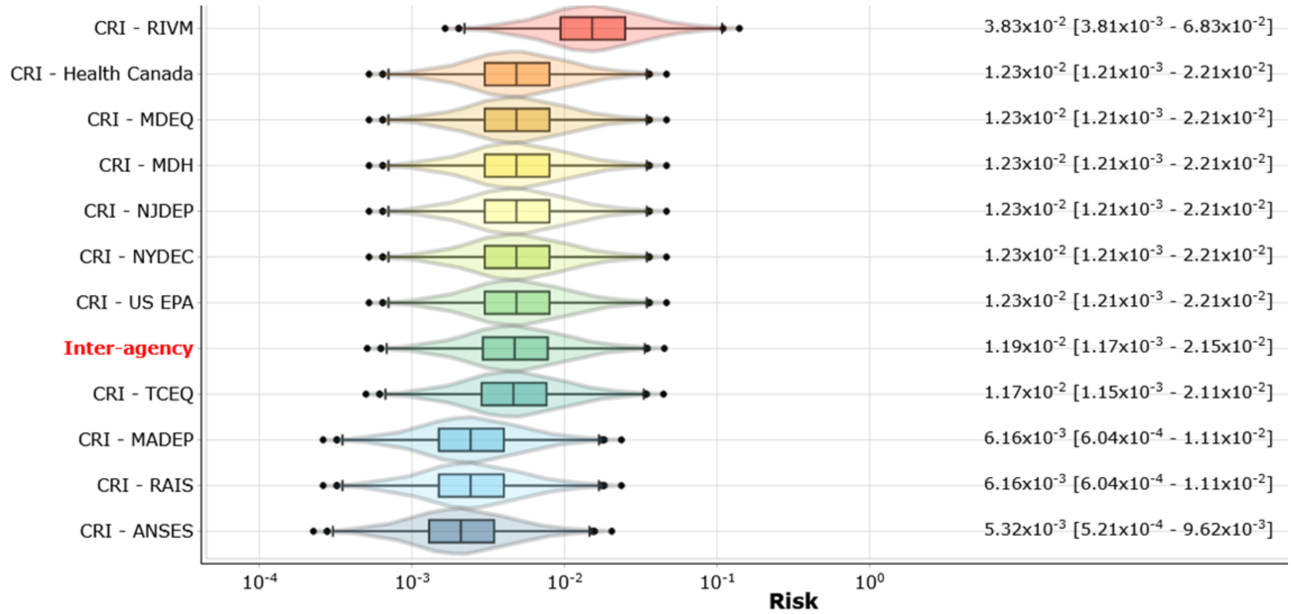

Fig.8: Distribution of the estimated cancer risk

Figure 7: Risk distribution according to health agencies

Exercise

Consider an individual exposed to benzene under the following exposure scenario:

• *Concentration*: geometric mean = 120 µg/m3 and geometric standard deviation = 2.4.

• *Exposure*: 12 hours per day, 200 days per year for 50 years.

• *Toxic effect*: leukemia.

• *Health agency*: OEHHA URE.

• *Exposed population*: 1,265 individuals.

Questions

1. What is this individual's cancer risk?

Answer: ELR = 3.18x10-3 [1.7x10-4 - 6.31x10-3]

2. Is this risk acceptable according to the recommendations of the U.S. Supreme Court?

Answer: No because ELR > 10-3

3. What is the probability of exceeding the risk value of 1x10-3?

Answer: approximately 46%

4. How many additional cancer cases are expected?

Answer: approximately 4 cases

5. What is the degree of agreement in the risk estimate between ANSES and Health Canada?

Answer: 76.3%

6. Which agency estimates the lowest risk? The highest?

Answer: Lowest: TCEQ (2.42x10-4; 0.31 cases ≈ 1 case). Highest: OEHHA (3.18x10-3; ≈ 4 cases)

The third tab (*Cancer – Chemicals*) provides information on cancer types and associated chemical substances, while the fourth tab provides occupational, environmental, and behavioral risk factors for the different cancers (Figure 8).

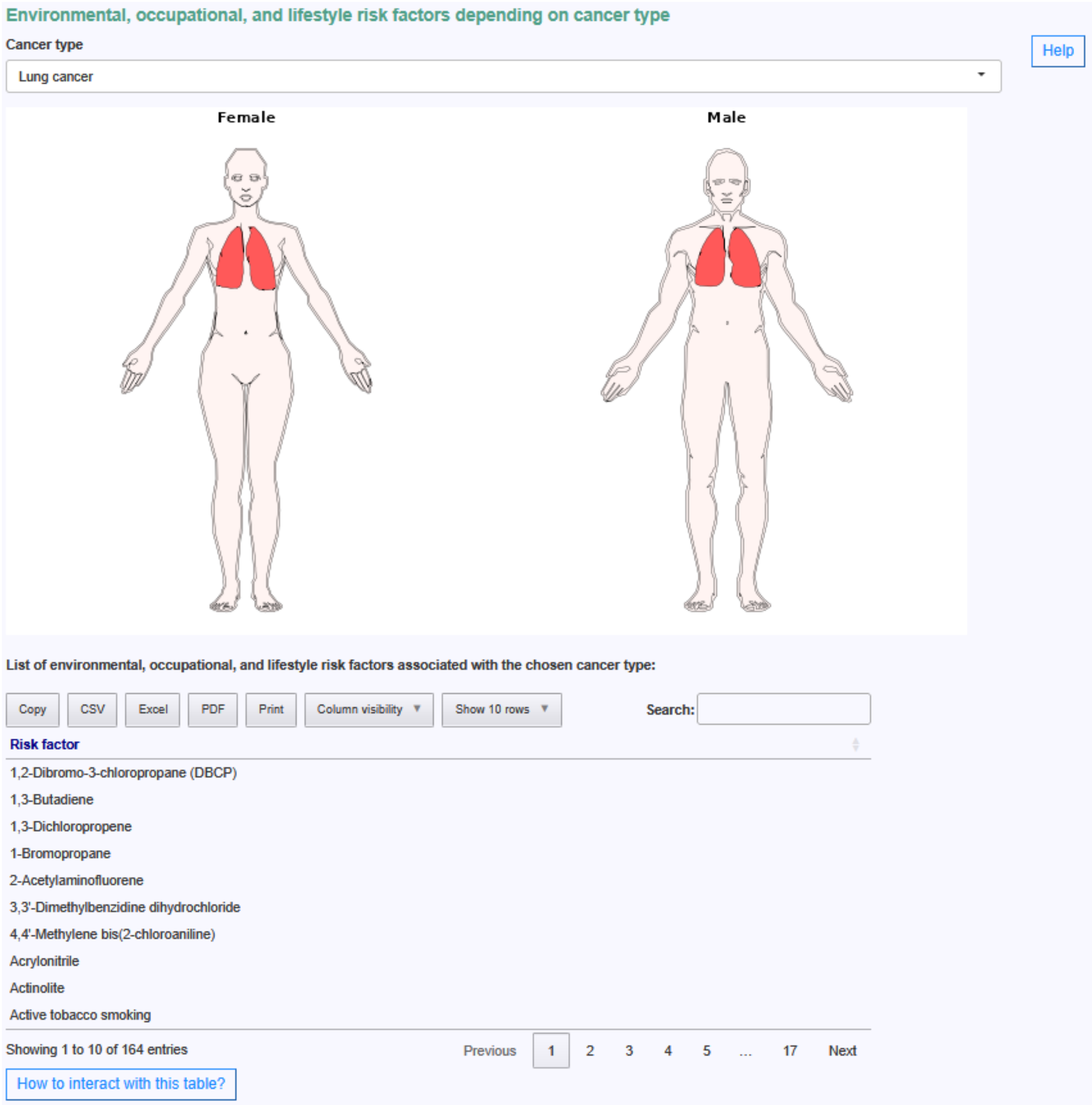

Figure 8: Environmental, occupational, and lifestyle risk factors for lung cancer

The fifth tab provides the data used (e.g., toxicological reference values) to model inhalation cancer risks. Finally, the last four tabs provide information on the use of *InCaRisk* in the scientific literature, available help, information about *InCaRisk*, and legal notices.

2.4 Simultaneous comparison of exposure scenarios

You can also compare multiple exposure scenarios simultaneously. To do so, select "*comparison of studies*" in the parameters panel (green box on the left side of the screen), under the "*study*" section. This option requires uploading a file containing at least two exposure scenarios (minimum 2 rows).

The figures and tables generated will be similar to those in the "*Inter-agency*" tab, but instead of comparing cancer risk estimates across different health agencies, they will compare different exposure scenarios.

Example files for illustration:

- Example_city PM2.5 concentration.csv
- Example_PM2.5 concentration by year.csv

3. Keywords and points to note

InCaRisk is a continuously evolving tool that is being enriched with new information and features. It is intended for researchers, prevention professionals, health agencies, and others who conduct risk assessments of cancer related to environmental and occupational exposures, by providing them with rich contextual information to support decision-making on prevention actions and measures. *InCaRisk* makes it possible to generate risk profiles based on exposure scenarios and offers the possibility of assessing the health impact of changes if preventive measures and/or behavioral changes were to occur.

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